

Samina Farheen

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EDUCATION

University of California, Berkeley

M.Eng. in Mechanical Engineering — Robotics & Autonomous Systems (Fung Scholar)

Berkeley, CA

Aug. 2026 – May 2027

University of California, Santa Barbara

B.S. in Mechanical Engineering — College of Engineering Honors - GPA: 3.7

Santa Barbara, CA

Sept. 2022 – June 2026

TECHNICAL SKILLS

Modeling: SolidWorks, Siemens NX, Fusion 360, GD&T, FEA, Arena PLM, Teamcenter

Hardware: Arduino, Raspberry Pi, Sensors (encoders, pressure, accelerometers, IMUs), Actuators (DC motors, servos, solenoids)

Programming: Python, C++ (Arduino IDE), MATLAB/Simulink, ROS 2, GitHub, Linux

PROJECTS

Autonomous Fluidic Morphing Gripper Arm | *M.Eng Graduate Capstone*

Aug. 2026 – Present

- Architecting 3 modular control systems using Python and ROS 2 for fluidic fingertip actuation, gripper movement, and arm trajectory estimation to enable full robot autonomy
- Implementing computer vision pipelines to classify object stiffness and autonomously deploy corresponding adaptive grasp profiles
- Establishing high-fidelity simulation models using MuJoCo to co-design hardware-software control architectures within strict 6-month timeline
- Integrating precise pressure and proximity sensing with a finite state machine framework to track task progression and maintain closed-loop feedback

VIO Drone Payload (VIPER) - Sponsored by Teledyne FLIR | *Senior Capstone*

Sept. 2025 – June 2026

- Directed a 10-member multidisciplinary team as project lead through full product lifecycle, managing timeline and sponsor technical reviews, to achieve only 1.5% inertial drift error during flight tests (1st place, UCSB Expo)
- Executed electromechanical system architecture design decisions, optimizing strict weight and form factor budgets for UAV application, with 300-gram mass and iPhone-scale sizing
- Spearheaded carbon fiber vibration-isolator mount design in SolidWorks via DFM/DFA principles for water-jet fabrication with ASME standard GD&T drawings and FEA optimization to mitigate high-frequency noise
- Conducted thermal benchmark testing, guided PCB layout and housing design to ensure tight electronic packaging and robust heat management for 45-minute flight

EXPERIENCE

ChargePoint, Inc.

Campbell, CA

Mechanical Design Intern

June 2025 – Sept. 2025

- Engineered an automated EV cycle counter (1-month turnaround), cutting bring-up time and saving nearly \$2,000 in labor costs
- Programmed custom C++ firmware on an Arduino Mega for 4 peripherals, configuring multi-slave SPI/I2C protocols for real-time logging
- Upgraded legacy hardware via precision soldering and rapid prototyping of custom mounting brackets and display panel
- Designed 2 DFM/DFA-compliant sheet metal assembly jigs in Siemens NX to streamline field charger installation
- Generated production-ready GD&T engineering drawings, managing via Arena PLM and NX Teamcenter, increasing hardware sprint efficiency by 20%

Hawkes Lab, UC Santa Barbara

Santa Barbara, CA

Undergraduate Research Assistant

Jan. 2025 – June 2026

- Redesigned 12-unit swarm robot fleet in Fusion 360, upgrading locomotion from 1 to 2 degrees of freedom
- Optimized robot-to-robot anchor interfaces to eliminate connection failure under dynamic load through 3 TPU 3-D print iterations
- Conducted force-based validation testing, processing datasets with Python and MATLAB to contribute to a graduate paper

NASA Ames Research Center

Mountain View, CA

Rotorcraft Aeromechanics Intern

June 2022 – Aug. 2022

- Modeled a 1:7 scale SolidWorks CAD assembly Side-by-Side test stand bed and gearbox, optimized for 3D printing
- Produced technical media for the HERO wildfire mitigation drone project, communicating key project metrics for internal departmental review